

Detonation Preliminaries

ZDP

I. Review of Shocks

Conservation Relations:

$$\rho_1 u_1 = \rho_0 u_0 \quad (\text{Mass});$$

$$p_0 + \rho_1 u_1^2 = p_0 + \rho_0 u_0^2 \quad (\text{Momentum});$$

$$h_1 + \frac{1}{2} u_1^2 = h_0 + \frac{1}{2} u_0^2 \quad (\text{Energy}; \quad \underbrace{h = e + \frac{p}{\rho}}_{\text{enthalpy}}).$$

Let $V \equiv \rho^{-1}$ (volume per unit mass), so that

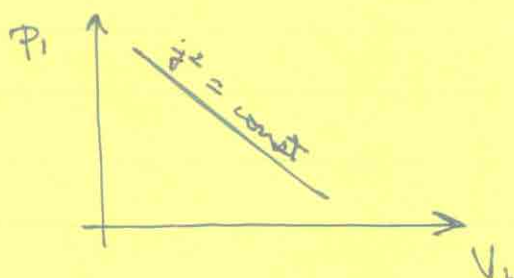
$$\frac{u_1}{V_1} = \frac{u_0}{V_0} = \underline{j}$$

$$p_1 + V_1 j^2 = p_0 + V_0 j^2$$

$$h_1 + \frac{1}{2} V_1^2 j^2 = h_0 + \frac{1}{2} V_0^2 j^2.$$

We have, from the momentum equation,

$$j^2 = - \frac{p_1 - p_0}{V_1 - V_0} = - \frac{\Delta p}{\Delta V} \Rightarrow p_1 = p_0 + (V_0 - V_1) j^2.$$



[This is sometimes called the Rayleigh line. I'll call it R.]

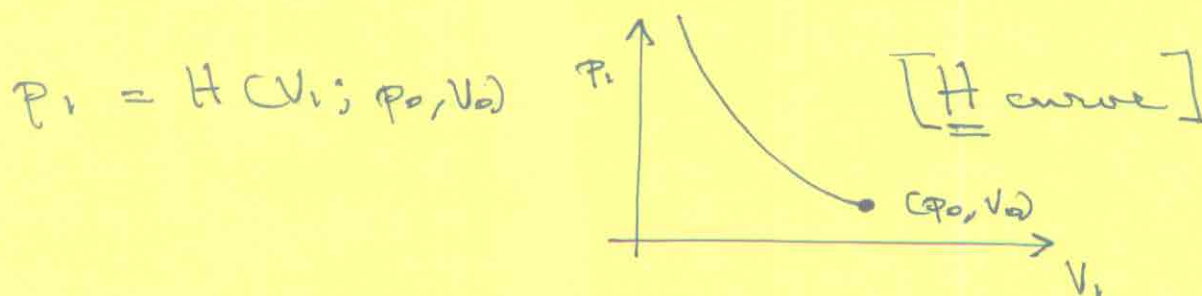
The energy equation becomes

$$\begin{aligned} h_1 - h_0 &= \frac{1}{2} (V_0^2 - V_1^2) j^2 &= \frac{1}{2} (V_0 - V_1) (V_0 + V_1) \left(\frac{P_0 - P_1}{V_0 - V_1} \right) \\ &= \frac{1}{2} (V_0 + V_1) (P_1 - P_0) ; \end{aligned}$$

or we can write

$$E_1 - E_0 = \frac{1}{2} (V_0 - V_1) (P_0 + P_1) .$$

For given functions $h_0(P_0, V_0)$ and $h_1(P_1, V_1)$ (or $E_0(P_0, V_0)$ and $E_1(P_1, V_1)$) we can solve for $P_1(P_0, V_0, V_1)$ to obtain the Hugoniot curve



If there are no chemical reactions and the thermodynamic properties of the fluid are unchanged across the shock, then H passes through (P_0, V_0) :

$$\left[\begin{array}{l} V_0 \rightarrow V_1 \Rightarrow E_1 \rightarrow E_0 \quad (\text{from } H \text{ relation}) \\ \Rightarrow P_1(E_1, V_1) \rightarrow \underbrace{P_1(E_0, V_0)}_{\substack{\text{because equation of state} \\ \text{unchanged across shock}}} \rightarrow P_0 \end{array} \right]$$

Weak shocks :

$$h = h(p, s)$$

$$\rightarrow @ s = \text{const. } (p_0, s_0)$$

$$\Rightarrow h_1 - h_0 = \partial_p h \Delta p + \frac{1}{2} \partial_{pp}^2 h \Delta p^2 + \frac{1}{6} \partial_{ppp}^3 h \Delta p^3 + \partial_s h \Delta s + O(\Delta p^4; \Delta s^2, \Delta s \Delta p, \dots)$$

$$\hookrightarrow @ p = \text{const. } (p_0, s_0)$$

$$\left[T ds = dE + p dV = dE + d(pV) - V dp = dh - V dp \right]$$

$$\Rightarrow dh = T ds + V dp$$

$$\Rightarrow \underline{\partial_s h = T}; \quad \underline{\partial_p h = V}$$

$$\Delta h = \cancel{V_0 \Delta p} + \frac{1}{2} \partial_{pp}^2 V \Delta p^2 + \frac{1}{6} \partial_{ppp}^3 V \Delta p^3 + T_0 \Delta s + \dots$$

$\rightarrow$ (dropping $O(\Delta s)$)

$$V = V(p, s) \Rightarrow \Delta V = \partial_p V \Delta p + \frac{1}{2} \partial_{pp}^2 V \Delta p^2 + \dots$$

$$\text{H curve: } \Delta h = \frac{1}{2} \Delta p [2V_0 + \partial_p V \Delta p + \frac{1}{2} \partial_{pp}^2 V \Delta p^2 + \dots]$$

$$= V_0 \Delta p + \frac{1}{2} \partial_{pp}^2 V \Delta p^2 + \frac{1}{4} \partial_{ppp}^3 V \Delta p^3 + \dots$$

$$\Rightarrow V_0 \Delta p + \frac{1}{2} \partial_{pp}^2 V \Delta p^2 + \frac{1}{6} \partial_{ppp}^3 V \Delta p^3 + T_0 \Delta s$$

$$= V_0 \Delta p + \frac{1}{2} \partial_{pp}^2 V \Delta p^2 + \frac{1}{4} \partial_{ppp}^3 V \Delta p^3$$

$$\Rightarrow T_0 \Delta s = \frac{1}{12} \partial_{ppp}^3 V|_s \Delta p^3.$$

For most fluids $\boxed{\partial_{ppp}^3 V|_s > 0}$. In a closed system, thermodynamic processes lead to $\Delta s > 0$ (second law).

So, if $\Delta S > 0$ and $\partial_p V|_s > 0$, then $\Delta \phi > 0$ for a weak shock. From the R relation, $j^2 = -\frac{\Delta \phi}{\Delta V} > 0$, we then find $\Delta V < 0$ in the case of a weak shock.

Note also:

$$\Delta V = \partial_p V|_s \Delta p + \dots$$

$$\frac{\partial^2 V}{\partial p^2}|_s = \frac{\partial}{\partial p} (\partial_p V)^{-1} = -(\partial_p V)^{-3} \frac{\partial^2 p}{\partial V^2}|_s$$

$$\Rightarrow \Delta \phi^3 = (\partial_p V|_s)^3 \Delta V^3$$

$$\begin{aligned} \Rightarrow T_0 \Delta S &= \frac{1}{12} \left[-(\partial_p V|_s)^3 \frac{\partial^2 p}{\partial V^2}|_s \right] (\partial_p V|_s)^3 \Delta V^3 \\ &= + \frac{1}{12} \frac{\partial^2 p}{\partial V^2}|_s (V_0 - V_1)^3 \end{aligned}$$

$$\Delta S, \partial_p V|_s > 0 \Rightarrow \underline{V_0 > V_1}.$$

We also find:



$$p = p(V, S) \Rightarrow \Delta p = \partial_p p|_s \Delta V + \frac{1}{2} \partial_p^2 p \Delta V^2 + \frac{1}{6} \partial_p^3 p \Delta V^3 + \partial_S p \Delta S$$

(general expansion)

$$\text{Adiabatic: } \Delta S = 0 \quad (\underline{A \text{ curve}}) \leftarrow$$

$$\text{H curve: } \Delta S = -\frac{1}{12} \partial_p^3 p|_s \Delta V^3 \leftarrow (\text{substitute into expansion})$$

(5)

Note that A and H have equal first and second derivatives at (p_0, V_0) . As we move from (p_0, V_0) , the difference between H and A is

$$H - A = - \frac{\partial p}{\partial s|_V} \cdot \frac{1}{2T_0} \frac{\partial^2 p}{\partial V^2|_S} \Delta V^2$$

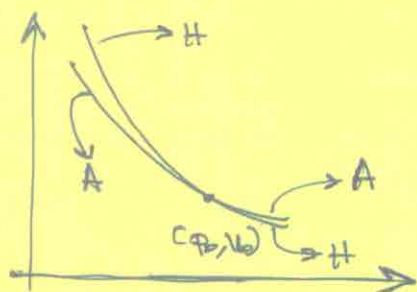
$$\left[\begin{array}{l} @ V = \text{const}; \quad \frac{\partial E}{\partial s|_V} > 0, \text{ and for nearly all fluids} \\ \frac{\partial p}{\partial E|_V} > 0 \Rightarrow \frac{\partial p}{\partial s|_V} > 0 \quad \left(\frac{\partial p}{\partial s|_V} = \frac{\partial p}{\partial E|_V} \frac{\partial E}{\partial s|_V} = T \frac{\partial p}{\partial E|_V} \right) \end{array} \right.$$

ideal gas: $E = C_V T = \frac{C_V}{R} p V$; $\frac{\partial p}{\partial E|_V} > 0$

$$\left[\begin{array}{l} \text{or } e^{S/C_V} = p V^{\gamma} \Rightarrow p \propto e^{S/C_V} @ s = \text{const} \\ \Rightarrow \frac{\partial p}{\partial s|_V} > 0 \end{array} \right.$$

$$\Rightarrow H - A = -1 < 0 > 0 < 0 > \Delta V^2 \propto -\Delta V^2$$

$$\frac{> 0 \text{ for } \Delta V < 0}{H \text{ above } A}; \quad \frac{< 0 \text{ for } \Delta V > 0}{H \text{ below } A}$$



$$\frac{\partial p}{\partial s|_V} > 0 \Rightarrow \frac{\partial s}{\partial p|_V} > 0$$

$\Rightarrow$ above (p_0, V_0) , H has a higher entropy than A ($\Delta S > 0$).

As a final point on weak shocks, note that

$$\left. \frac{\partial p}{\partial V} \right|_S (p_0, V_0) = \left. \frac{\partial p}{\partial V} \right|_H (p_0, V_0)$$

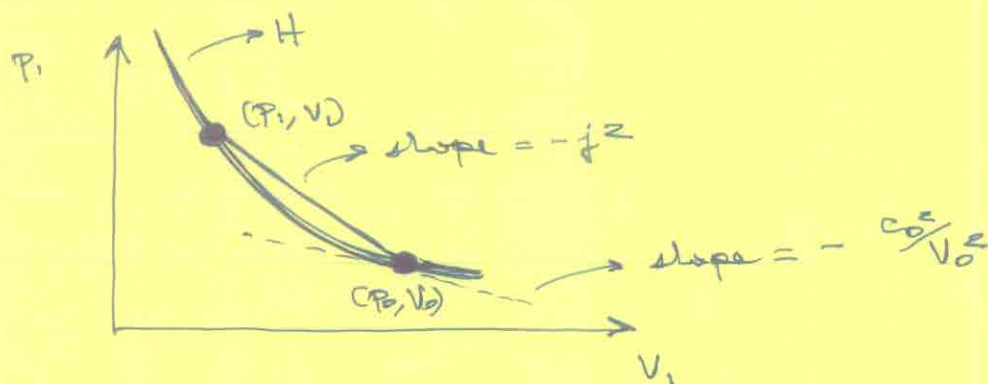
$\hookrightarrow$ tangent to A $\hookrightarrow$ tangent to H

$$c^2 = \left. \frac{\partial p}{\partial \rho} \right|_S = -V^2 \left. \frac{\partial p}{\partial V} \right|_S \Rightarrow -\frac{c^2}{V^2} = \left. \frac{\partial p}{\partial V} \right|_S.$$

Therefore, the slope of the tangent @ (p_0, V_0) is

$$\left. \frac{\partial p}{\partial V} \right|_H = -\underline{j^2} = -\frac{c_0^2}{V_0^2} \Rightarrow \underbrace{u_0 = D = c_0}_{\text{shock becomes acoustic}}.$$

As we move slightly up the H curve ($\Delta V < 0$, $|\Delta V| \ll V_0$) j^2 increases, and thus $j^2 > c_0^2/V_0^2 \Rightarrow \boxed{u_0 > c_0}$; i.e., the shock front moves supersonically wrt the upstream fluid.



The adiabat passing through (p_1, V_1) has a slope given by $-c_1^2/V_1^2$. Since the Taylor expansion of A and ~~shock~~ H agree to $O(\Delta V^2)$, we can just differentiate the expansion of A to find the slope of the adiabat at (p_1, V_1) :

$$2\phi|_{s_1} \approx 2\phi|_{s_0} + \partial_v^2 \phi|_{s_0} \Delta V.$$

Also,

$$j^2 = -\frac{\Delta \phi}{\Delta V} \approx -\left[2\phi|_{s_0} + \frac{1}{2} \partial_v^2 \phi|_{s_0} \Delta V\right]$$

$$\Rightarrow -\frac{j^2}{[c_1^2/V_1^2]} = \left(\frac{u_1}{c_1}\right)^2 = \frac{2\phi|_{s_0} + \frac{1}{2} \partial_v^2 \phi|_{s_0} \Delta V}{2\phi|_{s_0} + \partial_v^2 \phi|_{s_0} \Delta V}$$

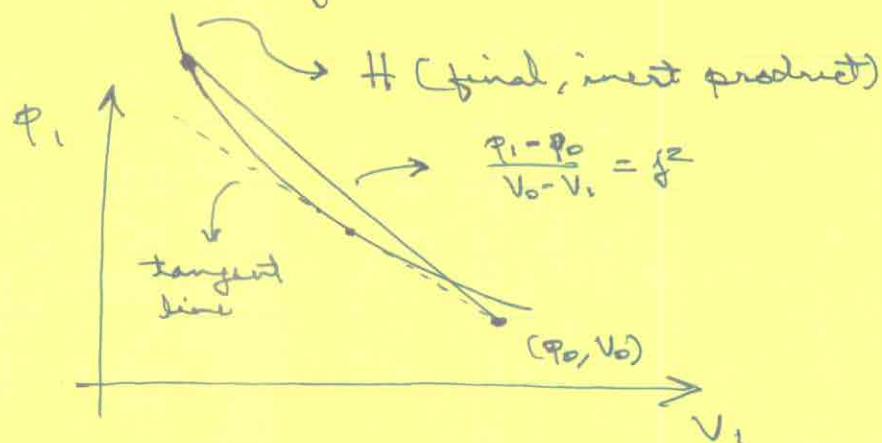
Since, $\partial_v^2 \phi|_{s_0} > 0$, the numerator is smaller than the denominator, and $[u_1 < c_1]$ i.e., the post-shock flow is subsonic.

In summary, for weak shocks:

$$\underline{\Delta S > 0}, \quad \underline{\partial_v^2 \phi|_s > 0} \Rightarrow \boxed{\begin{array}{l} p_1 > p_0, \quad V_1 < V_0 \\ u_0 > c_0, \quad u_1 < c_1 \end{array}}.$$

II. A More General Analysis (LL §87)

Allow for EOS change across thin shock/transition zone:



Dropping the subscript '1', we have

$$R: \quad \varphi = \varphi_0 + (V_0 - V)j^2$$

$$H: \quad h = h_0 + \frac{1}{2}(V_0^2 - V^2)j^2 \quad .$$

Assume that $\{\varphi_0, V_0, h_0\}$ are fixed parameters and $\{\varphi, V, h, j\}$ are (not independent) variables along H .

Differentiate R :

$$d\varphi = (V_0 - V)dj^2 - j^2 dV \quad .$$

Differentiate H :

$$dh = \frac{1}{2}(V_0^2 - V^2)dj^2 - j^2 V dV \quad .$$

$$1^{st} \text{ of thermo: } dh = Tds + Vd\varphi$$

$$\Rightarrow Tds + V \underbrace{(d\varphi + j^2 dV)}_{(V_0 - V)dj^2} = \frac{1}{2}(V_0^2 - V^2)dj^2$$

$$\begin{aligned} \Rightarrow Tds &= dj^2 \underbrace{\left\{ \frac{1}{2}(V_0^2 - V^2) - VV_0 + V^2 \right\}}_{= \frac{1}{2}V_0^2 + \frac{1}{2}V^2 - VV_0 = \frac{1}{2}(V - V_0)^2} \\ &= \frac{1}{2}(V - V_0)^2 dj^2 \quad . \end{aligned}$$

$$\Rightarrow \underline{\underline{Tds = \frac{1}{2}(V - V_0)^2 dj^2}} \quad .$$

since $\Delta V^2 > 0$, we see that $\frac{dj^2}{ds} > 0$ generally.

From the analysis of weak shocks, we know that s and j^2 increase above the tangent point ($\frac{dj^2}{ds} = 0$), and so the general relation $\frac{dj^2}{ds} > 0$ is consistent, but makes a stronger statement that applies everywhere along H .

Now consider $\frac{dj^2}{d\phi}$ along H :

$$d\phi + j^2 dW = (V_0 - V) dj^2$$

$$dW(\phi, s) = \partial_\phi V|_s d\phi + \partial_s V|_\phi ds$$

(since we already have $\frac{dj^2}{ds}$)

$$\Rightarrow d\phi + j^2 [\partial_\phi V d\phi + \partial_s V ds] = -\Delta V dj^2$$

$$\Rightarrow d\phi + j^2 \left\{ \partial_\phi V d\phi + \frac{\partial_s V}{2T} \Delta V^2 dj^2 \right\} = -\Delta V dj^2$$

$$\Rightarrow dj^2 \left[-\Delta V - j^2 \frac{\partial_s V}{2T} \Delta V^2 \right] = d\phi [1 + j^2 \partial_\phi V]$$

$$\Rightarrow \frac{dj^2}{d\phi} = \frac{[1 + j^2 \partial_\phi V]}{(-\Delta V) [1 + j^2 \frac{\partial_s V}{2T} \Delta V^2]} .$$

Tangency to H implies $\frac{dj^2}{d\phi} = 0$, and so

$$1 + j^2 \partial_\phi V|_s = 0 .$$

Note that

$$\left. \frac{\partial V}{\partial \phi} \right|_S = \left(\left. \frac{\partial p}{\partial V} \right|_S \right)^{-1} = - \frac{V^2}{c^2}$$

$$\Rightarrow 1 + j^2 \left. \frac{\partial p}{\partial V} \right|_S = 1 - \frac{u^2}{c^2}.$$

At the tangent point, $u = c$, just as in the weak shock case (no reactions) near (p_s, V_s) . In general, the tangent point is a sonic point (SP). Note that the SP on the H curve is a valid initial condition for a shock in the new fluid.

Along H, we also have

$$\left(\frac{dj^2}{d\phi} \right) / \left(\frac{dj^2}{dS} \right) = \frac{dS}{d\phi};$$

$$\frac{dj^2}{dS} > 0 \text{ generally} \Rightarrow \underline{\underline{\frac{dS}{d\phi} = 0}} \quad @ \quad \underline{\underline{SP}}.$$

From weak shocks, we know

$$\Delta S, \left. \frac{\partial p}{\partial V} \right|_S > 0 \Rightarrow \Delta p > 0, \Delta V < 0 \text{ above SP}$$

$$\Rightarrow \underline{\underline{\Delta S < 0}} \quad \underline{\underline{\text{below SP}}}.$$

From $1 + j^2 \partial_p V|_s = 1 - u^2/c^2$, we have

$$-\frac{d(u^2/c^2)}{dp} = \frac{dj^2}{dp} \partial_p V + j^2 \frac{d}{dp} \partial_p V ; \quad \frac{d \partial_p V(p,s)}{dp} = \partial_p^2 V|_s + \partial_p \partial_s V \frac{ds}{dp}$$

$$\Rightarrow -\frac{d(u^2/c^2)}{dp} = \frac{dj^2}{dp} \partial_p V + j^2 \left[\partial_p^2 V|_s + \partial_p \partial_s V \frac{ds}{dp} \right]$$

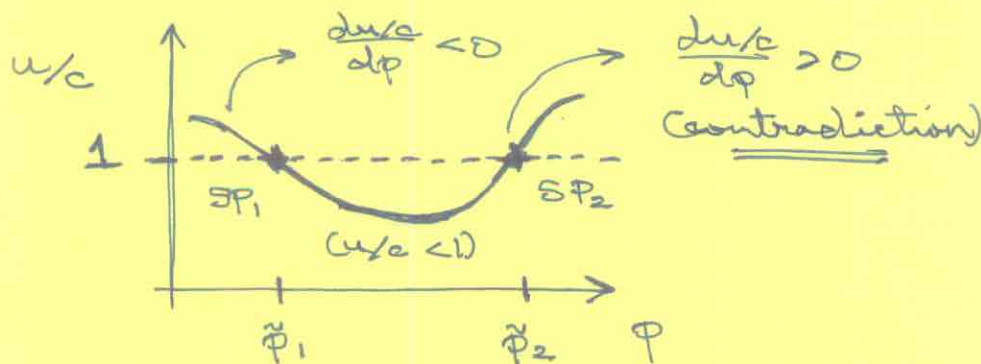
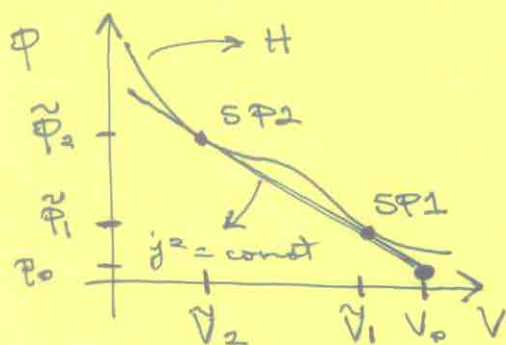
$$@ SP: \frac{dj^2}{dp} = \frac{ds}{dp} = 0 \Rightarrow \frac{d(u^2/c^2)}{dp} = -j^2 \partial_p^2 V|_s < 0$$

(since $\partial_p^2 V|_s > 0$)

$$\Rightarrow \frac{d(u^2/c^2)}{dp} < 0 @ SP \text{ generally.}$$

Above SP: $u/c < 1$; Below SP: $u/c > 1$.

If there is a second SP along H :



Since $\frac{du/c}{dp} < 0$ @ a SP (SP1, say), then the existence of another SP (SP2) means that $u/c < 1$ would have to increase ~~over some range~~ back to $u/c=1$, at which point $\frac{d(u/c)}{dp} \geq 0$, which contradicts our general finding.

Finally, we have the general conditions:

$$\Delta S, \partial_p^2 V|_S > 0 \Rightarrow$$

- $\Delta p > 0, \Delta V < 0$
- $u_0/c_0 > 1$
- $u_1/c_1 < 1$ above SP
 > 1 below SP
- only 1 SP
- f^2 has a single minimum @ SP.

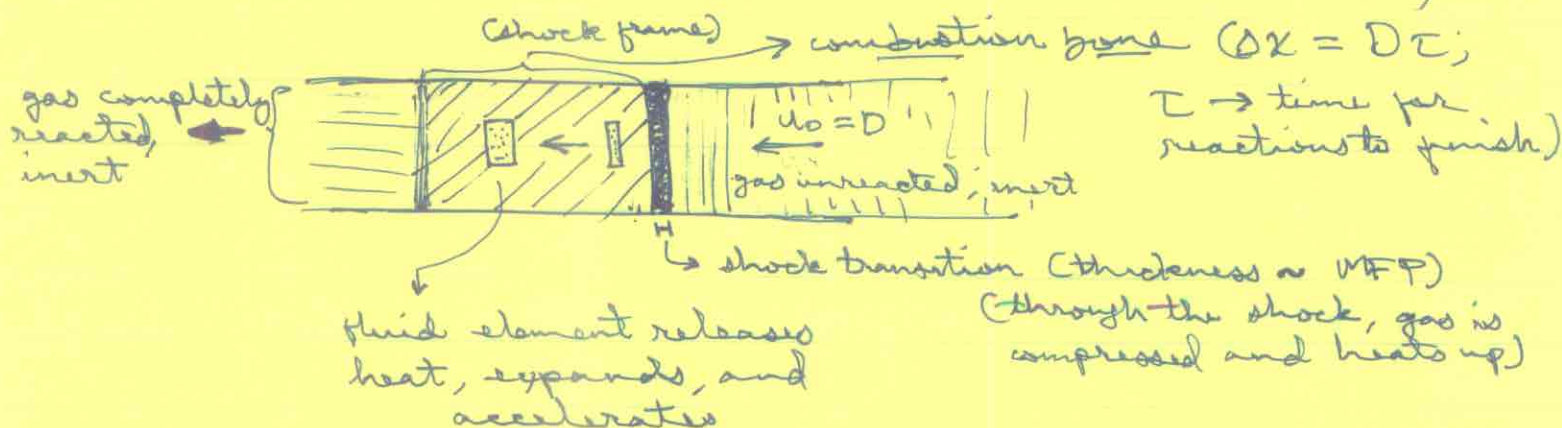
$$f^2_{\min} = \frac{D^2}{V_0^2} = \frac{u_1^2}{V_1^2} = \frac{c_1^2}{V_1^2}$$

$$\Rightarrow D^2 = \underbrace{\left(\frac{V_0}{V_1}\right)^2 c_1^2}_{\text{obtained from conservation relations}}$$

III. Detonation Basics

Detonation Wave:

$\tau \gg \text{MFP}/c_0$
 $\Rightarrow$ no significant reaction in shock.



In the simplest description we may treat the shock + combustion zone as a surface of discontinuity (the detonation wave), just as we treated shockwave without appealing to the microphysics within the transition zone.

Reaction Heat and Detonation H curve :

In a closed system, if we neglect internal friction and heat conduction,

$$0 = \Delta E + \Delta W.$$

If heat is released by chemical reactions, we have

$$\Delta E = E_f - E_i \quad ; \quad E_f = E_{of} + C_v T_f \quad ; \quad E_{oi} + C_v T_i$$

for an ideal gas. The constants are such that

$$E_{oi} - E_{of} = q \quad (\text{only difference matters})$$

$$\Rightarrow \Delta E = \underline{-q} + \Delta E_k \quad \searrow \text{difference in kinetic parts}$$

$$\Rightarrow q = \Delta E_k + \Delta W.$$

If the volume is held fixed, then $q = \Delta E_k$.

If $q > 0$, then $\Delta E_K > 0$, so that the chemical heat caused the molecules to move faster. It's as though we added $q > 0$ from an external reservoir ($TdS = q$) and defined $E \equiv E_K$.

By writing $E_0 = q + c_v T_0$, the latent chemical heat is equivalent to an introduction of positive entropy when $q > 0$. Moreover, we've ~~changed~~ changed the EOS ~~in~~ in a definite way, since q is absent from E_1 (post combustion).

As an example, let

$$h_0 = q + c_p T_0 ; \quad h_1 = c_p T_1 \quad (\text{no change in } c_p)$$

$$\Rightarrow c_p T_1 - q \left[1 + \frac{c_p T_0}{q} \right] = \frac{1}{2} (V_0 + V_1) (p_1 - p_0) ;$$

assume $q \gg c_p T_0$

$$\Rightarrow c_p T_1 - q \approx \frac{1}{2} (V_0 + V_1) \Delta p$$

$$c_p T_1 = \frac{\gamma}{\gamma-1} p_1 V_1$$

$$\Rightarrow \frac{\gamma}{\gamma-1} p_1 V_1 - q \approx \frac{1}{2} (V_0 + V_1) \Delta p$$

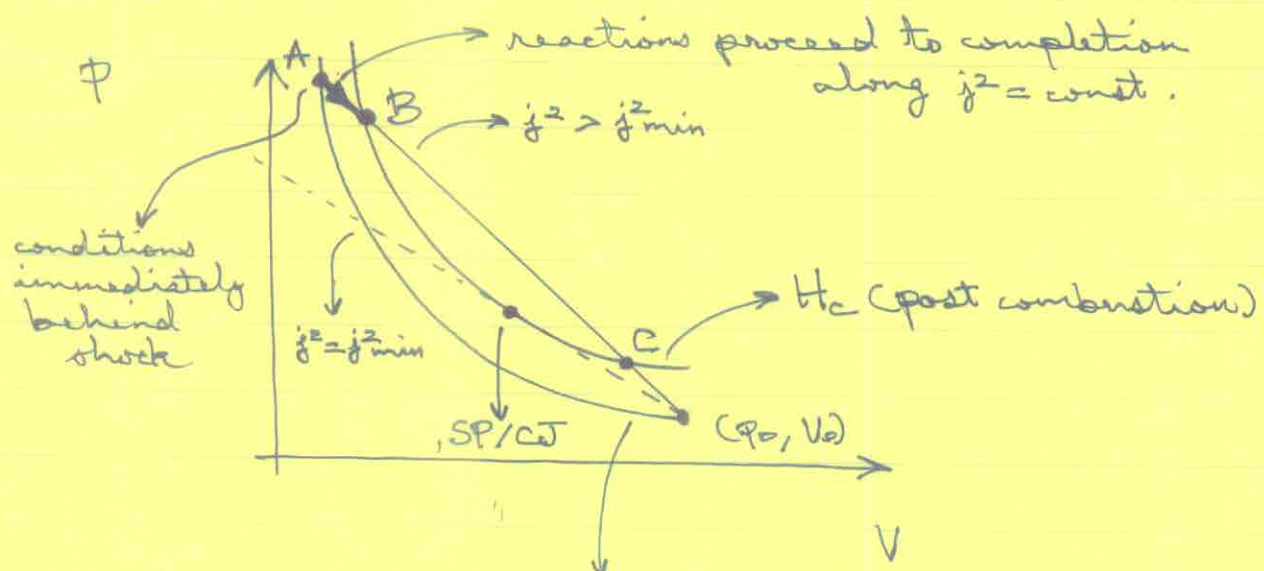
$$\text{let } V_1 \rightarrow V_0 \Rightarrow \frac{\gamma}{\gamma-1} p_1 V_0 - q \approx V_0 \Delta p$$

$$\Rightarrow p_1 \approx (\gamma-1) q / V_0 \neq p_0$$

(in fact $\gg p_0$ for $q \gg c_p T_0$)

In general, for $q > 0$, the H curve for the final reaction products sits above the H curve for the inert original mixture.

We now have the following picture for shock-induced combustion:



$j^2 = -\frac{\partial P}{\partial V}$ is a result of mass and momentum conservation, and has nothing to do with chemistry. Therefore, the state of the post-combustion gas must be a point on $R(j^2 = \text{const.})$ and H_c . Moreover, all intermediate states of the gas must sit on R .

Since H_c is the locus of possible final states, corresponding to the total release of chemical heat q , the transition from A to B must evidently map all possible intermediate states in the combustion zone. A smooth transition to the lower point C would

→ (which must then be removed to end up at C)

pass through states that demand additional release of chemical heat, more than is available. Since a discontinuous transition from B to C is out of the question (no additional shock would lead to a lower pressure), we must conclude that C does not represent a physical solution branch.

~~Now, for a pipe, the velocity of the detonation wave is given by~~

In order to satisfy some steady BC behind the detonation wave (e.g., vanishing velocity of the closed end of a pipe), in conjunction with the conserved fluxes, a rarefaction wave follows the detonation front. This is easy to see for a closed pipe. Following the combustion zone, the flow speed is given by

$$j^2 = u_B^2 / V_B^2 < c_B^2 / V_B^2.$$

This finite speed in the shock frame corresponds to speed $D - u_B$ in the lab frame. In order to match the no-flow boundary at the pipe end, the fluid speed must decrease further behind the combustion zone. This readjustment is accomplished by a rarefaction moving at the sound speed. The rarefaction appears because of the negative density wake that must accompany the post-shock density jump.

At point B, we know that $u_B < c_B$. Therefore, the forward moving rarefaction wave has a speed in the lab frame $D - w < c_B \Rightarrow w + c_B > D$.

The rarefaction wave overtakes the shock.

The result is a degradation of the shock until the zero-flow BC is satisfied.

The only steady solution that can satisfy all the BCs and conservation relation is the SP, corresponding to $j^2 = j^{\text{min}} = \frac{c_0^2}{V^2}$. In this case $u = c$ and $D = w + c$; the detonation wave just keeps ahead of the rarefaction wave. This is the Chapman-Jouguet solution.